SUBRAMANIAN B

Electronics and Communication Engineering Graduate

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SUMMARY

Electronics and Communication Engineering graduate with hands-on experience in Digital Electronics, Embedded Systems, FPGA, VLSI Design, Verilog HDL, and Computer Vision. Seeking a core engineering role to contribute to hardware design and embedded system development while continuously expanding technical expertise

EXPERIENCE

Embedded Systems Intern

Pantech E-Learning

- Developed embedded applications using microcontrollers
- Worked on hardware/software integration and real-time embedded solutions

The Institution of Electronics and Telecommunication Engineers (IETE)

- Gained hands-on experience in sensor integration, automation, and IoT-based application development

Lovely Offset Printers Pvt. Ltd., Sivakasi

- Worked on pre-press, press, and post-press operations, quality control, and process optimization

Madurai Railways

- Gained exposure to rail track maintenance, train control systems, and railway data networks

EDUCATION

B.E. ECE
Thiagarajar College of Engineering, Madurai – Anna University
CGPA 6.26 / 10.00

Diploma in Printing Technology
Sivakasi Institute of Printing Technology – DOTE
Score 93 / 100
- Gold Medalist

SSLC
Velammal Bodhi Campus, Theni – CBSE
Score 49.8 / 100.0

CONFERENCE

Clock Cycle Optimized Itoh-Tsujii Inversion Architecture for High-Performance ECC Applications
MQBIT'26 – International Conference on Materials for Electronics, Energy, Quantum and Biomedical Technologies

AWARDS

- **College Gold Medalist in Diploma**
- **2nd Prize, National Level Paper Presentation (2021)**
- **3rd Prize, National Level Paper Presentation (2022)**
- **1st Place, District Level Athletics (Running) Tournament**

KEY ACHIEVEMENTS

- **Hardware Optimization Success**
  Optimized hardware resource utilization by 30% in ECC applications.
- **Robot Efficiency Enhancement**
  Reduced obstacle detection latency by 25% in assistive robot.
- **Wi-Fi Booster Development**
  Improved signal booster range by 40% using antenna design.
- **Authentication System Completion**
  Achieved 95% accuracy in real-time vehicle authentication.

SKILLS

Programming
C | Embedded C | Java | Verilog HDL

Hardware
FPGA | ESP32 | ESP8266 | Arduino | STM32

EDA / Tools
MATLAB | Advanced Design System (ADS) | Xilinx ISE | STM32CubeIDE | Arduino IDE | PCB Artist | GNU Radio | Power BI

Core Areas
Digital Electronics | Analog Electronics | VLSI Design | Embedded Systems | Microcontrollers | PCB Design | Communication Systems

CERTIFICATIONS

MATLAB Onramp
MathWorks

Signal Processing Onramp
MathWorks

2-Day Workshop on EV Technologies
IIT Madras

CUSTOM

- **Part of VLSI Special Interest Group (SIG); Student Committee Member**
- **College People Leader; Cultural Secretary; ED Call Secretary**

PROJECTS

Efficient Hardware Implementation of Itoh-Tsujii Inversion Algorithm

- Designed an optimized hardware architecture for finite field inversion used in Elliptic Curve Cryptography (ECC)
- Reduced computational complexity, latency, and hardware resource utilization
- Implemented using Verilog HDL on a Virtex-4 FPGA in Xilinx ISE 14.7

Blind Assistance Robot using Computer Vision

- Developed an assistive robot using computer vision and AI to detect obstacles and recognize objects in real time
- Helped visually impaired individuals navigate surroundings more safely and independently
- Tools: YOLOv8, OpenCV, Google Colab, ESP32, Raspberry Pi 4 Model B

Smart Home Automation using ESP32

- Developed an IoT-based home automation system using ESP32, ESP8266, and ESP-NOW
- Implemented real-time appliance control and scheduled automation using NTP
- Tools: ESP32, ESP8266, ESP-NOW, Arduino IDE, Blynk IoT Platform

YOLOv8-based Two-Wheeler Authentication with Colour Detection

- Built a two-wheeler authentication system using YOLOv8 to detect vehicle make, model, and color
- Optimized the model for real-time performance with high accuracy in surveillance applications
- Tools: YOLOv8, Roboflow, Kaggle, Colab

Passive Wi-Fi Booster

- Designed a passive Wi-Fi signal booster using a PCB dipole antenna and cantenna for enhanced range
- Achieved a return loss of ~30.50 dB with an omnidirectional radiation pattern

Smart Irrigation System

- Built a smart irrigation system using real-time soil moisture sensors for efficient water usage
- Automated irrigation using live environmental feedback for sustainable cultivation
- Tools: Arduino Uno, Arduino IDE, Fritzing